

OSSP – PRATICAL

G HANSITA – 2520030332

SEC – 9A

WEEK 1

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root@Hansi: ~/OSSP/2520030332_Practical
GNU nano 7.2
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/types.h>
#include <sys/wait.h>

int main()
{
    char command[100];
    pid_t pid;

    printf("Enter a Linux command: ");
    scanf("%s", command);

    pid = fork();

    if (pid < 0)
    {
        perror("fork failed");
        return 1;
    }
    else if (pid == 0)
    {
        // Child process
        printf("Child Process ID (PID): %d\n", getpid());
        printf("Executing command: %s\n", command);

        execlp(command, command, (char *)NULL);

        perror("exec failed");
        exit(1);
    }
    else
    {
        // Parent process
        printf("Parent Process ID (PID): %d\n", getpid());
        printf("Parent is waiting for child process...\n");

        wait(NULL);

        printf("Child process completed.\n");
    }

    return 0;
}
```

```

root@Hansi: ~/OSSP/2520030332_Practical
root@Hansi:~# mkdir -p ~/OSSP/2520030332_Practical
root@Hansi:~# cd ~/OSSP/2520030332_Practical
root@Hansi:~/OSSP/2520030332_Practical# pwd
/root/OSSP/2520030332_Practical
root@Hansi:~/OSSP/2520030332_Practical# nano pratical_Q1.c
root@Hansi:~/OSSP/2520030332_Practical# ls
pratical_Q1.c
root@Hansi:~/OSSP/2520030332_Practical# ls -l
pratical_Q1.c
root@Hansi:~/OSSP/2520030332_Practical# gcc pratical_Q1.c -o pratical_Q1
root@Hansi:~/OSSP/2520030332_Practical# ./pratical_Q1
Enter a Linux command: ls
Parent Process ID (PID): 593
Parent is waiting for child process...
Child Process ID (PID): 594
Executing command: ls
pratical_Q1 pratical_Q1.c
Child process completed.
root@Hansi:~/OSSP/2520030332_Practical#

```

WEEK 3

```

root@Hansi: ~/OSSP/2520030332_Practical
GNU nano 7.2
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/wait.h>

int main()
{
    pid_t pid;

    pid = fork();

    if (pid > 0)
    {
        printf("Parent Process\n");
        printf("PID: %d\n", getpid());
        printf("PPID: %d\n", getppid());
    }
    else if (pid == 0)
    {
        printf("Child Process\n");
        printf("PID: %d\n", getpid());
        printf("PPID: %d\n", getppid());

        execl("/bin/ls", "ls", "-l", NULL);
    }
    else
    {
        perror("fork failed");
        return 1;
    }

    if (pid > 0)
    {
        wait(NULL);
        printf("Parent resumes after child finishes.\n");
    }

    return 0;
}

```

```

root@Hansi: ~/OSSP/2520030332_Practical
root@Hansi:~/OSSP/2520030332_Practical# nano pratical_Q3.c
root@Hansi:~/OSSP/2520030332_Practical# gcc pratical_Q3.c -o pratical_Q3
root@Hansi:~/OSSP/2520030332_Practical# ./pratical_Q3
Parent Process
PID: 626
PPID: 539
Child Process
PID: 627
PPID: 626
total 76
-rw-r--r-- 1 root root    55 Aug 17 07:08 destination.txt
-rwxr-xr-x 1 root root 16360 Aug 17 06:54 pratical_Q1
-rw-r--r-- 1 root root   839 Aug 17 06:54 pratical_Q1.c
-rw-r--r-- 1 root root   840 Aug 17 07:05 pratical_Q1.c.save
-rwxr-xr-x 1 root root 16224 Aug 17 07:08 pratical_Q2
-rw-r--r-- 1 root root  1080 Aug 17 07:07 pratical_Q2.c
-rwxr-xr-x 1 root root 16264 Aug 30 11:23 pratical_Q3
-rw-r--r-- 1 root root   665 Aug 30 11:23 pratical_Q3.c
-rw-r--r-- 1 root root    11 Aug 17 08:02 sample.txt
-rw-r--r-- 1 root root    55 Aug 17 07:07 source.txt
Parent resumes after child finishes.
root@Hansi:~/OSSP/2520030332_Practical# nano pratical_Q3.c
root@Hansi:~/OSSP/2520030332_Practical#

```

WEEK 4

```

root@Hansi: ~/OSSP/2520030332_Practical
GNU nano 7.2
#include <stdio.h>
#include <unistd.h>
#include <sys/wait.h>

int main()
{
    pid_t pid1, pid2;

    pid1 = fork();

    if (pid1 == 0)
    {
        printf("Child 1 Process\n");
        printf("PID: %d\n", getpid());
        sleep(2);
        return 0;
    }

    pid2 = fork();

    if (pid2 == 0)
    {
        printf("Child 2 Process\n");
        printf("PID: %d\n", getpid());
        sleep(3);
        return 0;
    }

    printf("Parent Process\n");
    printf("PID: %d\n", getpid());

    wait(NULL);
    printf("Parent collected one child using wait()\n");

    waitpid(pid2, NULL, 0);
    printf("Parent collected Child 2 using waitpid()\n");

    return 0;
}

```

```
root@Hansi: ~/OSSP/2520030332_Practical
root@Hansi:~/OSSP/2520030332_Practical# nano pratical_Q4.c
root@Hansi:~/OSSP/2520030332_Practical# gcc pratical_Q4.c -o pratical_Q4
root@Hansi:~/OSSP/2520030332_Practical# ./pratical_Q4
Child 1 Process
PID: 652
Parent Process
PID: 651
Child 2 Process
PID: 653
Parent collected one child using wait()
Parent collected Child 2 using waitpid()
root@Hansi:~/OSSP/2520030332_Practical# _
```